

Nabin Ranabhat

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nabin-ranabhat

Objective

Recent Computer Engineering graduate from Pulchowk Campus with hands-on experience in machine learning, NLP, and agentic AI systems. Skilled in fine-tuning transformer models, building data pipelines, and deploying ML workflows. Passionate about low-resource NLP, generative AI, and applying deep learning to real-world problems.

Education

Bachelor in Computer Engineering · *Pulchowk Engineering Campus*
Kathmandu, Nepal

- Aggregate: 82% — Relevant Coursework: Data Structures & Algorithms, DBMS, Artificial Intelligence, Generative AI (taught by Ekbana Solutions), DevOps Fundamentals, Software Engineering, Computer Organization & Architecture

Higher Secondary Education (+2 Science) · Gandaki Boarding School 2019 – 2021
Pokhara, Kaski

- GPA: 3.64 / 4.0

Internship

AI/ML Engineer Intern · *Proventus Technology Pvt. Ltd.* Nov 2025 – Jan 2026
Kathmandu, Nepal

- Contributed to **CratusPulse**, a patient-monitoring platform built to FCC and FDA-aligned compliance standards.
- Evaluated an ECG signal processing pipeline built on the **Pan-Tompkins++ algorithm** by benchmarking it against standard datasets including MIT-BIH Arrhythmia, MIT-BIH Noise Stress Test (NST), and LUDB.
- Analyzed detection performance metrics to validate pipeline reliability for regulatory-grade real-time vital-sign monitoring.

Technical Skills

Languages	Python, Java, C/C++, SQL, Rust
ML / DL Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn
NLP & Models	BERT, ByT5, mT5, FastText, Tokenizers, BLEU evaluation
Generative AI	LLM APIs (OpenRouter, Anthropic), Prompt Engineering, Agentic Systems, ChromaDB
Data & Pipelines	Pandas, NumPy, YouTube Data API, web scraping, data annotation
Backend & Deployment	Flask, Docker, GitHub Actions, REST APIs, Node.js
DevOps & Monitoring	Jenkins, Ansible, Prometheus, Grafana, Docker Compose
Databases	PostgreSQL, MongoDB, MySQL, Redis
Other	Git, Swagger/OpenAPI, Maven, Jupyter

Projects

Low-Resource Translation of Code-Mixed Romanized Nepali-English to Devanagari Script

Python, PyTorch, Hugging Face Transformers, ByT5, FastText, YouTube API, Pandas

- Developing a transliteration system to convert informal, code-mixed Romanized Nepali-English text into standardized Nepali Devanagari script — addressing a significant low-resource NLP gap.
- Scraped and cleaned a dataset of **24K+ YouTube comments**, manually annotating **2.8K+ parallel sentence pairs** for supervised training.
- Built a robust preprocessing pipeline using **phonetic normalization** and **FastText embeddings** to handle spelling inconsistencies and code-mixing noise.
- Fine-tuned the **ByT5 (byte-level T5)** transformer model for transliteration, achieving a **BLEU-4 score of 0.4996** and measurable character-level error reduction.
- Designed and prototyped an interactive chatbot interface to demonstrate real-time transliteration of code-mixed inputs.
- Currently expanding the dataset, experimenting with **mT5**, and improving grammatical correctness of outputs.

News Agent: Personalized News Digest System

Python, ChromaDB, OpenRouter API, Flask, Google News RSS

- Built a **multi-agent Python system** delivering personalized AI news digests via five specialized agents — Crawler, Embedding, Personalization, Writer, and Orchestrator — each with defined tool schemas and system prompts.
- Fetched live articles via **Google News RSS**; stored and retrieved them semantically using **ChromaDB** with a custom deterministic hash-based embedding function.
- Generated structured markdown digests via **OpenRouter LLM** with multi-model fallback (Mistral-7B, Gemma-2-9B); deployed a **Flask web UI** for end-to-end pipeline access.

BERT-NER Document Extraction System

BERT, NLP, Python, TensorFlow

- Developed a system using **BERT for Named Entity Recognition (NER)** to automatically extract information from citizenship documents and auto-fill forms, streamlining document processing workflows.
- Built a **browser extension** that works on websites supporting document image uploads, enabling one-click extraction and automatic population of relevant form fields.
- **GitHub:** [GitHub Link](#)

Automated CI/CD Pipeline for Node.js Application

Jenkins, Docker, Ansible, Prometheus, Grafana, GitHub

- Built a fully automated CI/CD pipeline integrating GitHub, Jenkins, Docker, Ansible, Prometheus, and Grafana into a cohesive DevOps workflow on Ubuntu 22.04.
- Configured Jenkins to trigger on every GitHub push, building a versioned Docker image, running smoke tests, and pushing to Docker Hub.
- Automated zero-touch deployment via an **Ansible** playbook; instrumented the app with **prom-client** exposing HTTP counters, latency histograms, and Node.js runtime metrics.
- Provisioned **Grafana** dashboards visualizing request rate, p99 latency, heap usage, CPU, and memory; reduced pipeline cycle time from ~120s to ~70s via Docker layer caching.

Certifications

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| • Samsung Innovation Campus: Artificial Intelligence | May 2025 – Nov 2025 |
| • Artificial Intelligence Fellowship — fusemachines | Apr 2025 – Oct 2025 |
| • Samsung Innovation Campus: Coding and Programming | Dec 2023 – Apr 2024 |

Problem Solving

Solved 150+ problems on LeetCode covering arrays, dynamic programming, graphs, trees, and backtracking.

Profile: leetcode.com/u/FlamboyantFate

Languages

Nepali — Native **English** — Professional proficiency **Hindi** — Conversational